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In studying the impact of shelf space changes on unit sales, space elasticity was hypothesized to be a function of several product-specific variables, including physical properties, merchandising characteristics, and use characteristics. The model was tested using stepwise multiple regression, and it was found that the impact of shelf space on unit sales was very small relative to the effects of other variables.

The Relationship Between Shelf Space and Unit Sales in Supermarkets

INTRODUCTION

Retailers generally believe that display exposure influences product unit sales, so shelf space allocation is regularly manipulated to increase sales and profits. For example, products with high gross margins frequently are displayed in supermarkets at eye level, at high-traffic locations, and with many facings. However, little empirical analysis of results has been reported, and practically no attempt has been made to generalize the impact of changes in shelf space on unit sales to classes of products.

Based on the belief that general patterns of shelf space-sales relationships can be identified, this research investigated the impact of shelf space upon unit sales for a large number of grocery products. A model relating the effects of changes in space (space elasticity) to physical properties, merchandising characteristics, and use characteristics of particular products was proposed and tested.

SPACE ELASTICITY

Space elasticity is defined as the ratio of relative change in unit sales to relative change in shelf space:

$$E = \frac{(U_{t_1} - U_{t_0})/U_{t_0}}{(S_{t_1} - S_{t_0})/S_{t_0}}$$

where U is unit sales and S is shelf space.

Lee [17] proposed a general model for space elasticity in which unit sales increase at a decreasing rate as shelf space is increased. Although (as in most of the literature) units of shelf facings were not specified, total display area was used as the measure of shelf space. Brown and Tucker [2] distinguished curves for three classes of products: (1) unresponsive products, usually also price inelastic, (2) general use products, for which increasing shelf space gives diminishing returns, and (3) occasional purchase products which respond slowly to shelf space increases until a large display causes the sales curve to rise steeply to a point of diminishing returns. This last pattern implies a step-function or threshold effect for shelf space. Although others [4, 12] have speculated about such an effect operating for certain products, there is little empirical evidence of the nature of this phenomenon.

Also, estimates of the relationship between shelf space and sales may be influenced by factors other than shelf space [3], and shelf space changes for a given product affect its sales as well as those of a range of substitute and complementary items. This consideration and others led to the conclusion that the impact of changes in shelf space on unit sales may be indeterminate at low levels of allocation, curvilinear at intermediate levels, and constant (zero) at high levels. To the extent that unit sales are responsive to changes in shelf space at all, the relationship should be direct, and except for random aberrations, space elasticity should be positive.

Variations in Elasticity

The relationship between shelf space and unit sales is not uniform across products, periods, and stores. Combined sales have been found to increase monotonically as shelf space was increased, and, likewise, pooled

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Table 1
INDEPENDENT VARIABLES IN THE MODEL

Variable	*Hypothesized relationship to space elasticity*	*Operational measure*
Physical properties		
x_1 Package size	Positive	Square inches of display area
Merchandising characteristics		
x_2 Retail price	Inverse	Dollars per unit
x_3 Brand type	Greater for private brands	National or private/packer brands
x_4 Market share	Inverse	SAMI Denver market brand share
x_5 Rate of sales	Positive	Average base period unit sales per thousand customer transactions
x_6 Shelf capacity	Inverse	Multiple of base period average weekly unit sales
x_7 Merchandise variety	Positive	Number of products in merchandise category
x_8 Availability of substitutes	Positive	Sum of: (1) the number of other brands, (2) other varieties of the same brand, and (3) other sizes of the same brand
Use characteristics		
x_9 Repurchase frequency	Inverse	Likelihood of purchase on a given shopping trip
x_{10} Extent of unplanned purchasing	Positive	DuPont index of unplanned purchasing
x_{11} Immediacy of use	Positive	Assumed primary use: (1) requires considerable preparation, (2) requires minimum preparation, or (3) ready to use

sales were found to decrease when shelf facings were reduced, although individual products differed widely in their response to changes in shelf space [1, 21].

The effect of shelf space also varies by geographic regions and by stores within regions [15], as well as by store size [11]. Moreover, apparent negative elasticities frequently have been observed [1, 15, 21]. Since there is no rational explanation for this phenomenon, it may be attributed to inadequacies in experimental control.

Cox [8] found a weak linear relationship between shelf space and unit sales for all product classes examined over a wide range of space allocations, and Bates [1] found a linear relationship for most space-elastic products, with a curvilinear model holding for the others. In both studies, the range of facings tested was greater than that expected under usual operating conditions for such products in typical supermarkets.

In large, well run supermarkets, most items are allocated enough shelf space to position them relatively far along their respective shelf space/unit sales curve to a point of diminishing marginal return. Under this condition, one would expect few items to have so little shelf space that large unit sales could be realized from nominal space increases. Indeed, zealousness in allocating space to private brands could put these items beyond the point where any incremental return would be realized from additional space. This implies that greater elasticity may be expected for a shelf space decrease than for an equivalent space increase.

Although space elasticity may vary among grocery products, usual store stocking practices suggest a linear relationship between shelf space and unit sales. In the following section, the independent variables in the proposed model are described (see Table 1).

THE RESEARCH VARIABLES

Package size. In previous research, greater space elasticity with respect to package size has been found for: smaller size cans of vegetables (but not fruits) [21], "regular" sizes of liquid salad dressing and vinegar [20], smaller packages among "regular" size products in a variety of categories (although greater unit sales changes were realized for larger packages because of larger absolute space increments) [1], and larger, difficult-to-stock items [5]. Perhaps these findings have been confounded by other variables, especially rate of sales, which is generally higher for smaller packages. Allowing for this factor, it was hypothesized that space elasticity is positively related to package size.

Retail price. Bates [1] compared items which responded to space changes with those which did not and found that the responsive group were lower priced. This finding is consistent with consumer risk-taking theory, which holds that consumers are more likely to respond to promotional efforts (including shelf space changes) for low-priced items because of the relatively low risk involved. Therefore, it was hypothesized that space elasticity is inversely related to retail price.

Brand type. Oliver [21] found private brand products more space elastic than their national-brand counterparts, although evidence to the contrary has been reported [1]. Because shelf space display likely is a less important component of the merchandising mix for national brands than for private label brands, it was hypothesized that private or packer's brands would have greater space elasticity than equivalent national brands.

Market share. The brand-type argument was extended to brand reputation in general. It was hypothesized that items with larger market share will be less space-elastic.

Rate of sales. Faster-selling items have been shown to respond more to changes in shelf space than slower selling items. Although this finding may be a consequence of the difficulty of detecting elasticity for slower-selling products, it was hypothesized that a space elasticity is positively related to rate of sales.

Shelf capacity. *Progressive Grocer* [5, 19] reported that customers purchased 20% more when shelves were well stocked than they did under "normal" conditions; however, the methodological weakness of these studies should be noted. Shelf-space elasticity was hypothesized to be greater for products whose rates of sales are large in relation to shelf capacity.

Merchandise variety. Items in multibrand categories have been found to exhibit greater space elasticity than items in categories having fewer brands [22]. Bates reasoned that [1, p. 59]:

> When there are a large number of items in a merchandise category, the chances for distinction are substantially reduced and the importance of visibility in determining sales should be increased. However, if there are only a few items in the merchandise category, shelf space should exert a smaller influence on sales because there are only a few items to serve as distractions, even when shelf space is limited.

Thus, it was hypothesized that space elasticity is positively related to the number of items in a product category.

Availability of substitutes. Bates also found that items with a lot of competition from substitutes were more responsive to shelf-space changes than those without competition [1], but determination of substitutes is difficult. Pauli and Hoecker [22] concluded that canned vegetables should be considered substitutes beyond their types—for example, that a brand of peas competes with all canned vegetables. At the extreme, this argument could include the alternatives of prepared take-home food and restaurant meals.

Despite problems in devising an operational measure of substitution, it was hypothesized that space elasticity is positively related to the number of substitutes for a given product.

Repurchase frequency. Kuehn [16] found that brand loyalty decreased with the length of interpurchase time. If brand names are less important for infrequently purchased products, it may be argued that other components of the merchandising mix, such as shelf space display, are more important. Thus it was hypothesized that space elasticity is inversely related to frequency of purchase.

Extent of unplanned purchasing. Unit sales of "impulse" items are said to be affected by changes in shelf space more than sales of staple items. There is no really good measure of impulse buying, but the one most widely used compares consumer purchase intentions with shopping outcomes [23], although measured intentions probably understate actual intentions [10]. Cox [7, 8] and Bates [1] found impulse products more space-elastic than staple products. It was hypothesized that space elasticity is positively related to the degree of impulse purchasing.

In an additional measure of impulse devised for this study, items were coded according to preparation time: (1) considerable preparation, (2) minimum preparation, or (3) ready to use.

The way in which these variables were operationalized is given in Table 1.

THE MODEL

These hypothesized relationships led to specification of a general equation for product space elasticity, as follows:

$$E_{i,r,h,n,p} = a + b_1x_1 + b_2x_2 + b_3x_3 \cdots b_{11}x_{11} + e$$

where E_i is the estimated coefficient of space elasticity for a given item, $x_1 - x_{11}$ are the independent variables, a and $b_1 - b_{11}$ are constants, and the following factors are held essentially constant:

r = retail price
h = shelf height
n = in-store location
p = promotion.

Also, e = error term, a random variable with an assumed mean of zero; b_1, b_3, b_5, b_7, b_8, b_{10}, and b_{11} are expected to be positive and the other values of b negative; and x_3 is one for national brands and zero for private brands.

METHODOLOGY

About 500 grocery products were studied under actual operating conditions, in which shelf space was increased or decreased for test items. Unit sales were observed for 5 to 12 weeks before and after changes. Shelf space changes were made according to recommendations of: (1) store management or (2) COSMOS, a computerized management information system which monitors the profitability of grocery items per unit of shelf space and calculates shelf space allocations using programmed heuristics.[1]

Four large supermarkets of a regional chain, which already had experimental COSMOS installations, served as the test stores; 24 others in the chain in the same area

[1] In this analysis, changes based on management recommendations and COSMOS were pooled, although this is judged not to have distorted the findings. These two influences are considered separately in other research [9].

were used as control stores. The test stores were large, modern, well managed, and profitable. They average 25,000 square feet, somewhat larger than the typical supermarket. Sales averaged over $100,000 per store, per week, about twice the national average for supermarkets. Sales per square foot of sales area averaged $6.50 per week, also substantially higher than the national average.

Items whose retail price, shelf level, display, or promotion changed were eliminated from the sample, although in some instances only for that portion of the period affected. Test items' unit sales were adjusted for seasonal and promotional variations by a factor comparing control stores' base-period unit sales with test-period unit sales for each product:

$$U_{t_1} = U_{\text{observed } t_1}(U_{\text{control } t_0}/U_{\text{control } t_1}).$$

A few items with extremely high or low estimated coefficients of space elasticity were excluded from the sample because they were probably affected by undetected promotional activity, generally of item substitutes or complements.

Finally, because unit sales were calculated from warehouse withdrawal records rather than audits, the test periods were required to last until unit sales became a multiple of shelf capacity sufficient to minimize the effect of any variation.[2]

Collectively, these restrictions reduced an original sample of 955 items to a final sample of 493 items.

RESULTS

Stepwise multiple regression was performed using the independent variables to explain space elasticity. First, in testing for multicolinearity, only two correlations as high as .5 were observed: (1) the DuPont index of unplanned purchasing [23] with availability of substitutes (−.49) and (2) the index with retail price (−.53), both significant at the .001 level. The correlation of measures of unplanned purchasing (the DuPont index and immediacy of use) was .33 ($p < .001$), so both variables were retained because of their contribution to explaining dimensions of impulse.

The regression analysis [20] yielded an R^2 of .032, which, when adjusted for degrees of freedom [18], was .012. This result implies that collectively the independent variables explained only one percent of the variance in space elasticity. This low figure, combined with a relatively large standard error of the estimate and standard errors of the regression coefficients, indicates that the hypothesized independent variables has little power to predict the dependent variable in the given circumstances.

Despite this failure, analysis suggested that the hypothesized variables might more accurately predict elasticity for subsets of products: rate of sales, extent of change in display area, test store, and merchandise category. The results of these analyses are reported below and summarized in Table 2.

Rate of sales. It was noted before that space elasticity should be greater for faster-selling items. To test this hypothesis, two groups of test items were delineated:

	Number of items	*Percentage*	$\bar{R}^2$
Rate of sales more than 3 units per 1,000 customer transactions (about 40 units) per week	160	32	.065
Fewer than 3 units	333	68	.003
Total	493	100	

Although $\bar{R}^2$ was indeed higher for items having greater sales, neither $\bar{R}^2$ was consequential.

Change of display area. Regression analysis was performed for those cases in which display area was increased and decreased:

	Number of items	*Percentage*	$\bar{R}^2$
Display area increased	287	58	.020
Display area decreased	206	42	.066
Total	493	100	

Although the independent variables explained more variance in elasticity for decreases than for increases, neither $\bar{R}^2$ was consequential.

Test stores. Similar regression analysis was conducted for each of the four test stores. Despite some variation in $\bar{R}^2$'s, the differences were not attributable to store differences:

Store	*Number of items*	*Percentage*	$\bar{R}^2$
1	121	25	.130
2	116	24	0
3	105	21	.080
4	151	30	.020
Total	493	100	

Merchandise categtory. Only canned vegetables had enough test items to permit meaningful regression analysis. Canned vegetables were compared with all other items:

	Number of items	*Percentage*	$\bar{R}^2$
Canned vegetables	124	25	.165
All other items	369	75	.005
Total	493	100	

Analysis of regression results for canned vegetables

[2] Limits were set so that an inventory variance of one-third of shelf capacity for a given product would bias imputed rate of movement by no more than 15%. Unit sales in this study exceeded by 18 times the assumed likely variation between beginning and ending inventory of one-third shelf capacity, so the average bias under these assumed constraints was 5.5%.

Table 2

OBSERVED MEAN SPACE ELASTICITY FOR SAMPLE SUBSETS

Variable	Elasticity		N	Agreement with hypothesized relationship
	Mean	Difference[a]		
Package size				
Less than 20 square inches	.233	.050	286	No
Greater than 20 square inches	.183		207	
Retail price				
Less than 50¢	.222	.038	359	Yes
Greater than 50¢	.184		134	
Brand type				
National brands	.176	.118[b]	342	Yes
Private and packer brands	.294		151	
Market share				
SAMI share greater than .25	.186	.065	298	Yes
SAMI share less than .25	.251		195	
Absolute sales level				
Base unit sales less than 3	.206	.019	333	Yes
Base unit sales greater than 3	.225		160	
Shelf capacity				
Less than 1.25 weeks' sales	.207	.013	301	No
Greater than 1.25 weeks' sales	.220		192	
Merchandising variety				
Less than 30 items in category	.219	.020	328	No
More than 30 items in category	.199		165	
Availability of substitutes				
Less than 5 items	.211	.005	352	Yes
More than 5 items	.216		141	
Repurchase frequency				
Likelihood less than .25	.211	.002	313	No
Likelihood greater than .25	.213		180	
Extent of unplanned purchasing				
DuPont index less than .5	.115	.127[b]	116	Yes
DuPont index greater than .5	.242		377	
Immediacy of use				
Requires preparation	.199	.027	234	Yes
Ready to use	.226		259	
Change of display area				
Decreased space	.132	.138[b]	206	No
Increased space	.270		287	
Store				
1	.209		121	
2	.201		116	
3	.244		105	
4	.201		151	
Merchandise category				
Canned vegetables	.191	.083	369	
All other items	.274		124	

[a] Difference between mean observed values.
[b] Significant at the .25 level or better.

revealed the same weaknesses noted for other samples—especially high standard errors of the estimate and the regression coefficients, relative to the estimated coefficients.

Analysis of Space Elasticity by Variable

Although regression analysis failed to support the hypothesis that space elasticity could be explained by the independent variables, some of them could be related significantly to space elasticity. To test for such relationships, the sample was divided according to high and low values, based on the distribution of values, for each variable, and the resultant mean value for space elasticity was compared for each subset. These results are shown in Table 2. Variances for observed elasticity generally were large relative to the subset means. Consequently, relatively large differences between the means of the high and low groups frequently were nonsignificant statistically, even at the .25 level. Significant differences were found in the direction hypothesized for brand type and extent of unplanned purchasing. In addition, elasticity was found to differ significantly, but not in the direction hypothesized, for display area increases vs. decreases. Further breakdown into smaller subsets failed to reveal any additional meaningful patterns.

SUMMARY

A model proposed to explain space elasticity as a function of several product-specific variables was tested under actual operating conditions for nearly 500 grocery products. Unfortunately, it was not able satisfactorily to explain observed variations in space elasticity. The most important reason for the model's failure to perform as hypothesized was that for most products under typical operating conditions the impact of changes in shelf space on unit sales is very small relative to the effects of other variables. Despite controlling for effects of other variables by excluding suspect items from the final "refined" sample, unrecognized and uncontrolled interrelationships among item substitutes and complements (including promotions, price changes, and item additions and deletions) must have affected test-item unit sales. The observation of apparent negative elasticities in one-third of the cases is probably a manifestation of this difficulty.

In this study, space elasticity averaged .212 for all items, showing a positive relationship between shelf space and unit sales. Although not all products have been found to be space elastic [1, 7, 8, 21], previous research also supports this general conclusion [1, 22]. As in other studies, measurement of the relationship was complicated by effects of numerous factors other than shelf space, causing variations of observed elasticities.

What are the implications to the retailer of generally positive space elasticity of the magnitude observed? The typical experimental treatment in this study involved a shelf space change (increase or decrease) of approximately 40%. Elasticity averaged slightly more than .2, implying that on the average, unit sales changed about 8% in the direction of shelf space changes.

From the retailers' point of view, an 8% change has tremendous profit implications. However, since space-sensitive items could not be identified in advance by the variables tested in this research, the opportunity to influence unit sales selectively is limited. Profit might be maximized by increasing display space for *all* items, although the payoff is doubtful [9, p. 246].

Even if the payoff were more favorable, it is unlikely that this strategy would work. Relative changes of display space may be more significant than absolute increases, and increasing shelf space for every grocery item is analagous to building a store too large for its market. The complaints of supermarket operators caught in this predicament attest to that strategy's unprofitability.

At the retail level, operational considerations should take precedence over merchandising considerations. High labor costs require minimization of restocking costs and avoidance of stock-outs before particular products are favored with additional space. These conditions, aggravated by a continuous stream of new items, leave the typical store with limited opportunity for discretionary manipulation of shelf space. However, the findings of this study confirm the general industry practice of allocating such space to private label and impulse products.

If it is valid to assume that supermarkets commonly allocate shelf space sufficient to place items in the "most likely zone," the finding that elasticity associated with shelf space increases was double that for decreases (Table 2) suggests that space for individual products could be increased and then cut back systematically. Where manufacturers' representatives are given discretion to manipulate shelf space, they might employ this strategy.

CONCLUSIONS AND IMPLICATIONS FOR FUTURE RESEARCH

This study has shown the obvious need for a good instrument to measure impulse. More discriminating independent variables might be proposed to explain space elasticity, including the degree to which elasticity for a given product varies by store and display location, although they are unlikely materially to improve explanation.

Although these and other issues probably warrant investigation, this research supports previous opinion that important payoffs are unlikely from investigations of space elasticity. Clearly, shelf space is a less important component of the marketing mix than hitherto has been assumed. Failure to explain elasticity makes it unlikely that specific product space elasticities can be "factored" into merchandising management systems in any simple way. Given these findings, simulation of the merchandising effects of shelf space reallocations on sales and profits is likely to remain impractical.

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